Henrique Soares Assumpção e Silva

Machine Learning Researcher
Belo Horizonte, Brazil

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Website Github
Linkedin Google Scholar

Education

MS in Computer Science Mar 2025 – Jul 2026
Federal University of Minas Gerais (UFMG), Brazil *GPA: 99%*

- **Thesis**: Semidefinite Optimization on Laplacians of Highly Regular Graphs
- **Advisor**: Prof. Gabriel Coutinho

BS in Computer Science Mar 2020 – Dec 2024
Federal University of Minas Gerais (UFMG), Brazil *GPA: 94%*

- **Thesis**: Algebras, groups and graphs
- **Advisor**: Prof. Gabriel Coutinho
- **Minor**: Mathematics

Professional Experience

Machine Learning Research Scientist Jun 2025 - Ongoing
Inter, Brazil *Full Time*

- Lead an interdisciplinary team on the *Large Transaction Models* project, directing the pre-training and optimization of 10B+ parameter foundation models on large-scale financial datasets across multi-node GPU clusters for credit risk analysis and downstream financial forecasting.
- Created, open-sourced, and led the development of CodeEvolve, an evolutionary coding agent for scientific discovery. CodeEvolve outperforms other state-of-the-art frameworks, including DeepMind's AlphaEvolve, on multiple code generation benchmarks.

Machine Learning & Optimization Research Fellow Dec 2020 - May 2025
Computer Science Department - UFMG, Brazil *Fellowship*

- Developed and deployed a Variational Autoencoder in PyTorch for predictive maintenance on industrial machinery, improving accuracy over legacy systems by 10%.
- Designed a recurrent neural network to predict emotional shifts across large online communities, outperforming established baselines by 20%.
- Derived novel semidefinite programming bounds for NP-hard combinatorial problems using spectral graph theory.

Data Science Instructor Jun 2022 - Dec 2022
Usiminas, Brazil *Internship*

- Designed and delivered hands-on instruction in Python, Pandas, and PyTorch to engineering teams at a major steel manufacturer.
- Mentored internal teams building ML tools, contributing to the automation of legacy data pipelines.

Machine Learning Research Intern Aug 2021 - Jun 2022
Inter, Brazil *Internship*

- Developed *DELATOR*, a Graph Neural Network framework for modeling large financial transaction networks with 20M+ nodes and 100M+ edges.
- Deployed the pipeline into production banking environments, improving money laundering detection over legacy baselines.

A.I. Research & Development Engineer Mar 2021 - Aug 2021
Plus Three, USA *Internship*

- Built NLP pipelines using Transformer-based models for multi-turn question answering and text generation in production web systems.
- Wrote a technical advisory series on algorithmic bias and fairness for the non-profit *AIandYou*.

Papers

Conference Proceedings

- **Henrique Assumpção**, Diego Ferreira, Leandro Campos, Fabricio Murai. CodeEvolve: an open-source evolutionary coding agent for algorithmic optimization. *Findings of the Association for Computational Linguistics: EMNLP 2026*.
- **Henrique Assumpção**, Fabrício Souza, Leandro Lacerda Campos, Vinícius T. de Castro Pires, Paulo M. Laurentys de Almeida, Fabricio Murai. DELATOR: Money Laundering Detection via Multi-Task Learning on Large Transaction Graphs. *2022 IEEE International Conference on Big Data (Big Data)*.

Journal Articles

- **Henrique Assumpção**, Gabriel Coutinho, Chris Godsil. Total Conformal Rigidity in Graphs. Under revision at *Journal of Graph Theory*, 2026.
- **Henrique Assumpção**, Gabriel Coutinho. Semidefinite programming bounds on fractional cut-cover and maximum 2-SAT for highly regular graphs. *Mathematics of Operations Research*, 2026.
- Bárbara Silveira*, **Henrique Assumpção***, Fabricio Murai, Ana Paula C. da Silva. Predicting user emotional tone in mental disorder online communities. *Future Generation Computer Systems*, 2021.

*Equal contribution.

Talks and Presentations

- CodeEvolve: Evolutionary Agents for Scientific Discovery. Invited Talk. AIDDA 2026. Remote.
- CodeEvolve: an Open Source Evolutionary Coding Agent for Algorithmic Discovery and Optimization. Invited Talk. Inter Science Talks 2026. Belo Horizonte, Brazil.
- Gauge duality for parameters of highly regular graphs. Invited Talk. ILAS 2026. Blacksburg, USA.
- Evolutionary Coding Agents. Invited Talk. Automated Discovery at Scale 2026. San Francisco, USA.
- MAXCUT, Association Schemes and Semidefinite Programs. Invited Talk. CNMAC 2025. Rio de Janeiro, Brazil.
- Primal-dual inequalities for the approximate maximum-cut value of Graphs in Homogeneous Coherent Configurations. Poster Presentation. CNMAC 2024. Porto de Galinhas, Brazil.
- DELATOR: Money Laundering Detection via Multi-Task Learning on Large Transaction Graphs. Oral Presentation. IEEE BigData 2022. Osaka, Japan.

Technical Skills

Programming Languages: C, C++, CUDA, Python
Tools & Technologies: Apache Spark, AWS Sagemaker, Docker, Git, HuggingFace, JAX, Megatron-LM, Pytorch
Languages: Portuguese (Native), English (C2), Spanish (B2), French (B1)